MATH3603

UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
Probability Theory
Test 2

1. Consider the graph in Figure 1 where each node represents a city and each edge (i, j) is associated with a weight w_{ij}. Person A initially resides in city 1 (node 1) and moves from one city to another in the following manner. If this month the person resides in city i, then the next month he will reside in one of its neighbor city j with probability proportional to w_{ij}. In other words, the one-step transition probability from i to j is given by

$$P_{ij} = \frac{w_{ij}}{\sum_k w_{ik}}$$

where $w_{ij} = 0$ if (i, j) is not an edge.

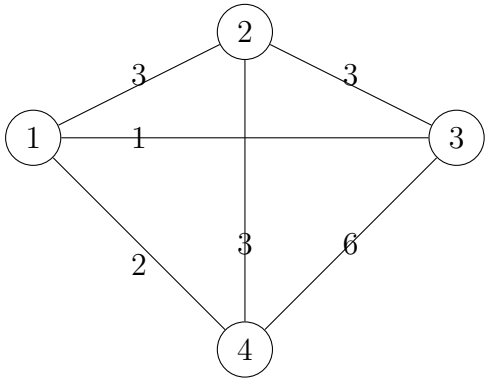


Figure 1: $w_{12} = w_{21} = 3$, $w_{23} = w_{32} = 3$, $w_{14} = w_{41} = 2$, $w_{34} = w_{43} = 6$, $w_{24} = w_{42} = 3$, $w_{13} = w_{31} = 1$

(a) Let $X_n = i$ if person A resides in city i during month n. Write the one-step transition probability matrix P for the Markov chain $\{X_n, n \geq 0\}$.

Solution.

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{6} & \frac{1}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{10} & \frac{3}{10} & 0 & \frac{3}{5} \\ \frac{2}{11} & \frac{3}{11} & \frac{6}{11} & 0 \end{pmatrix}$$

□

(b) Find $(x_1; x_2; x_3; x_4) \in \mathbb{R}^4$ such that

$$\begin{pmatrix} x_1 & x_2 & x_3 & x_4 \end{pmatrix} P = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 \end{pmatrix}. \tag{1}$$

Solution. We have

$$\sum_{k=1}^{4} w_{1k} = 6, \ \sum_{k=1}^{4} w_{2k} = 9, \ \sum_{k=1}^{4} w_{3k} = 10, \ \sum_{k=1}^{4} w_{4k} = 11$$

Then

$$\begin{pmatrix} x_1 & x_2 & x_3 & x_4 \end{pmatrix} = \begin{pmatrix} \frac{1}{6} & \frac{1}{4} & \frac{5}{18} & \frac{11}{36} \end{pmatrix}$$

is a solution of (1). □

(c) Recall that $X_0 = 1$, i.e., person A initially resides in city 1. Find the probability that city 2 is chosen before city 4;

Solution. We consider the Markov chain $\{Y_n, n \geq 0\}$ with state space $\{1, 2, 3, 4\}$ and transition probability matrix

$$\begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{6} & \frac{1}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{10} & \frac{3}{10} & 0 & \frac{3}{5} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Then states 1, 2 and 3 are transient and state 4 is recurrent (absorbing). And starting from node 1, node 2 is visited before node 4 in the Markov chain $\{X_n, n \geq 0\}$ if and only if starting from state 1, state 2 is ever visited in the Markov chain $\{Y_n, n \geq 0\}$. To compute the probability that the Markov chain $\{Y_n, n \geq 0\}$ ever makes a transition into state 2 given that it starts in state 1, we consider the matrix

$$P_T = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{6} \\ \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{1}{10} & \frac{3}{10} & 0 \end{pmatrix},$$

which specifies only the transition probabilities between transient states 1, 2 and 3. Define:

$$S = (I - P_T)^{-1} = \begin{pmatrix} 1 & -\frac{1}{2} & -\frac{1}{6} \\ -\frac{1}{3} & 1 & -\frac{1}{3} \\ -\frac{1}{10} & -\frac{3}{10} & 1 \end{pmatrix}^{-1}.$$

Then the desired probability $f_{1,2}$ can be computed by

$$f_{1,2} = \frac{s_{1,2}}{s_{2,2}} \simeq \frac{0.8049}{1.4390} = 0.5593.$$

□

(d) Let

$$N = \min\{n \geq 1 : X_n > 2\}.$$

In other words, N denotes the number of transitions until the person resides in city 3 or in city 4. Compute $\mathbb{E}[N]$.

Solution. We consider the Markov chain $\{Z_n, n \geq 0\}$ with state space $\{1, 2, 3, 4\}$ and transition probability matrix

$$\begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{6} & \frac{1}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

The transient states of the chain $\{Z_n, n \geq 0\}$ are state 1 and state 2. Note that the mean number of transitions until either state 3 or state 4 is entered in the Markov chain $\{X_n, n \geq 0\}$ is equal to the mean number of time periods that the Markov chain $\{Z_n, n \geq 0\}$ is in state 1 or state 2. Consider the matrix

$$P_T = \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{3} & 0 \end{pmatrix}$$

Then,

$$S = (I - P_T)^{-1} = \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{3} & 0 \end{pmatrix}^{-1} = \begin{pmatrix} 1.2 & 0.6 \\ 0.4 & 1.2 \end{pmatrix}.$$

Then by Proposition 28, the mean number of times periods that $\{Z_n, n \geq 0\}$ is in state 1 or state 2, starting from state 1 is $1.2 + 0.6 = 1.8$. Hence

$$\mathbb{E}[N] = 1.8.$$

□

(e) Each month, person B who resides in city i will choose a neighbor city j to visit, after which he decides to whether move to city j for the next month or stay in the same city i for the next month. Suppose that residing in city i, person B will choose to visit a neighbor city j with probability proportional to w_{ij} as given in Figure 1. Denote by $\alpha_{i,j}$ the probability that person B decides to move from i to j after visiting city j. Give the values of $\alpha_{i,j}$ for each $i \neq j$ so that in the long run, person B resides in the four cities with equal frequency.

Solution.

$$\alpha_{i,j} = \min\left(\frac{\sum_k w_{ik}}{\sum_k w_{jk}}, 1\right) = \min\left(\frac{x_i}{x_j}, 1\right), \quad i \neq j,$$

where (x_1, x_2, x_3, x_4) is the solution of (1). □

2. Suppose that the number of accidents N per week at an industrial plant is a Poisson random variable with mean 4. Denote by Y_i the number of workers injured in ith accident. Suppose that $\{Y_1, \ldots, Y_i, \ldots\}$ are i.i.d. geometric random variables with mean 2, and are independent with N. Denote by S the total number of workers injured in one week:

$$S = \sum_{i=1}^{N} Y_i$$

Note that if $N = 0$ then $S = 0$.

(a) What is $\mathbb{E}[S]$?

Solution. $\mathbb{E}[S] = \mathbb{E}[N]\mathbb{E}[Y_1] = 8.$ □

(b) What is $\mathrm{Var}(S)$?

Solution. We know that

$$\mathrm{Var}(Y_1) = 2, \quad \mathrm{Var}(N) = 4.$$

Hence by Proposition 22 in Lecture 7,

$$\mathrm{Var}(S) = 2^2 * \mathrm{Var}(N) + 2 * \mathbb{E}[N] = 4 * 4 + 2 * 4 = 24.$$

□

(c) What is $\mathbb{P}(S=0)$?

Solution.

$$\mathbb{P}(S=0) = \mathbb{P}(N=0) = e^{-4}.$$

□

(d) What is $\mathbb{P}(S=1)$?

Solution.

$$\mathbb{P}(S=1) = \mathbb{P}(N=1)\mathbb{P}(Y_1=1) = e^{-4} * 4 * \frac{1}{2} \simeq 0.03663.$$

□

(e) What is $\mathbb{P}(S=2)$?

Solution. We apply Proposition 23 in Lecture 11 with $\alpha_1 = \frac{1}{2}$ and $\alpha_2 = \frac{1}{4}$,

$$\mathbb{P}(S=2) = 2\left(\frac{1}{2}\mathbb{P}(S=1) + \frac{2}{4}\mathbb{P}(S=0)\right) = \mathbb{P}(S=1) + \mathbb{P}(S=0) \simeq 0.0549.$$

□

(f) What is $\mathbb{P}(S=4)$?

Solution. We apply Proposition 23 in Lecture 11 with $\alpha_1 = \frac{1}{2}$, $\alpha_2 = \frac{1}{4}$, $\alpha_3 = \frac{1}{2^3}$, $\alpha_4 = \frac{1}{2^4}$ and $\alpha_5 = \frac{1}{2^5}$.

$$\mathbb{P}(S=3) = \frac{4}{3}\left(\alpha_1\mathbb{P}(S=2) + 2\alpha_2\mathbb{P}(S=1) + 3\alpha_3\mathbb{P}(S=0)\right) = \frac{23}{6}e^{-4} \simeq 0.0702.$$

$$\begin{aligned}\mathbb{P}(S=4) &= \frac{4}{4}\left(\alpha_1\mathbb{P}(S=3) + 2\alpha_2\mathbb{P}(S=2) + 3\alpha_3\mathbb{P}(S=1) + 4\alpha_4\mathbb{P}(S=0)\right) \\ &= \frac{53}{12}e^{-4} \simeq 0.08089.\end{aligned}$$

□